

II B.Tech II Semester Regular Examinations, Apr/May 2009
SOFTWARE ENGINEERING
(Common to Computer Science & Engineering, Information Technology
and Computer Science & Systems Engineering)

Time: 3 hours

Max Marks: 80

Answer any FIVE Questions
All Questions carry equal marks

1. (a) List the application of software engineering.
(b) Explain about capability Maturity Model Integration. [6+10]
2. (a) What are the various framework activities of the concurrent development process model.
(b) Explain the modeling activity and the deployment activity of the concurrent development process model with the state-transition diagrams. [8+8]
3. (a) Define Ethnography. What are the requirements that are associated with ethnography?
(b) Is ethnography combined with prototyping? Discuss the statement. [8+8]
4. (a) What type of design classes does the designer create? Explain about a “Well formed” design class?
(b) Explain the refinement procedure of an analysis class into a design class with an example. [8+8]
5. (a) Explain with example, how is the data hiding and data encapsulation implemented in object oriented languages.
(b) Explain how is polymorphism used during software designing phase.
(c) Write a program to demonstrate the concept of class scope and method scope. Explain the lifetime of a variable. [6+5+5]
6. Why do we need metrics for design model? Describe in detail the architectural design metrics. [16]
7. Write short notes on:
 - (a) Direct metrics
 - (b) Indirect metrics
 - (c) Risk component drivers
 - (d) DRE [4+4+4+4]
8. (a) Define software quality.
(b) What are the points that software quality emphasis on.
(c) What are the activities of SQA? [3+3+10]

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1. (a) How software engineering is different from hardware engineering?
(b) Distinguish between personal process model and team process model. [8+8]
2. (a) Explain the classical life-cycle process model with a real-time scenario.
(b) Give the reasons and situations where this model fails. [6+10]
3. (a) Define a scenario. Write a sample use case scenario for an article downloading in the library system.
(b) Write a short notes on interviewing technique. [8+8]
4. (a) Give a good and bad software design for any particular problem.
(b) What is data design? Explain how it is done with examples. [16]
5. What are the steps in interface design? Explain them with an example. [16]
6. Compare metrics for OO design & architectural design. [16]
7. (a) Explain about risk identification.
(b) Explain about risk projection. [8+8]
8. (a) What are the steps to be followed by statistical quality assurance?
(b) Explain with an example data collection for statistical software quality assurance.
(c) Explain the six sigma strategy for statistical software quality assurance. [4+6+6]

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1. (a) What is a legacy software?
(b) What type of changes is made to legacy systems if it exhibits poor quality?
(c) Explain why legacy systems evolve as time passes. [3+5+8]
2. (a) Differentiate between prototyping and incremental process models.
(b) Illustrate RAD process model with an example. [8+8]
3. (a) What are the goals of requirements engineering process?
(b) With the help of a neat diagram depict the requirement engineering process activities.
(c) Explain the spiral model of requirements engineering process. [2+6+8]
4. What are the major elements of the design model? Explain the abstraction dimensions and process dimensions of the analysis model and design model. [16]
5. (a) What is meant prototype? Why is it necessary to create prototype? Discuss about the different categories of prototype.
(b) Describe the process that are involved in unified approach for software development. [8+8]
6. Differentiate between class oriented metrics & OO metrics. [16]
7. Write short notes on:
(a) direct
(b) indirect
(c) correctness
(d) maintainability [4×4=16]
8. Discuss about Formal Technical Review. [16]

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1. What is a process pattern? Describe the template of a process pattern with an example. [16]
2. (a) Discuss the problems of using natural language for defined user and system requirements.
(b) Explain how structuring natural language into forms can help avoiding some of the difficulties of natural language. [8+8]
3. What is the purpose of domain analysis? Explain how it is related to the concept of requirements patterns. [16]
4. What are the major elements of the design model? Explain the abstraction dimensions and process dimensions of the analysis model and design model. [16]
5. Explain in detail the terms
 - (a) Task elaboration
 - (b) Object elaboration
 - (c) Work flow analysis [6+5+5]
6. Differentiate between class oriented metrics & OO metrics. [16]
7. Write short notes on:
 - (a) directed metrics
 - (b) indirect metrics
 - (c) public metrics
 - (d) private metrics [16]
8. (a) What is the need of software reviews?
(b) What are the different types of reviews?
(c) Which is the effective method of software review & explain. [3+5+8]
